

OFFLINE ADAPTATION

COLLECT

Driving demos

RGB + task + state
teacher vehicle motion

VALIDATE

Training corpus

406 episodes - 93,733 samples
388 task strings

ADAPT

Fine-tune SmolVLA

60k steps - batch 8
30-step SE(2) targets

FREEZE

v6-60K checkpoint

fixed weights for evaluation

DEPLOY FROZEN WEIGHTS

10 Hz CLOSED-LOOP DEPLOYMENT

SENSE

Observation

front RGB + instruction
speed - yaw rate - 0

INFER

Frozen SmolVLA

30-step SE(2) chunk
per queue refill

BUFFER

Action queue

10 Hz replay
five-step overlap

TRANSFORM

ROS 2 bridge

speed cap + slew
curvature limit

ACTUATE

Gazebo vehicle

vehicle motion
camera + state

CAMERA + STATE FEEDBACK